KAIZEN MATHS

Lesson starter

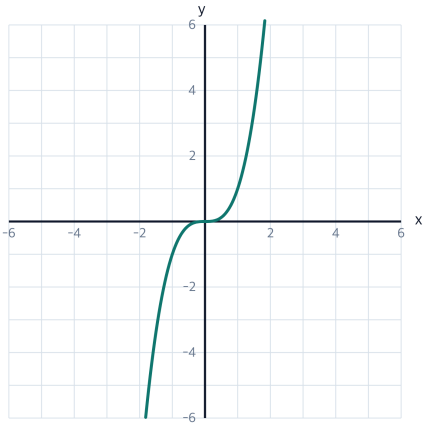
Graph Transformations and Curve Sketching

Parent Graphs

Name _____

SA1

Identify each parent graph and state one key feature.



Identify the parent graph shown and state one key feature.

ALG-146D-L1 · SA

Show your working

KAIZEN MATHS

Lesson starter

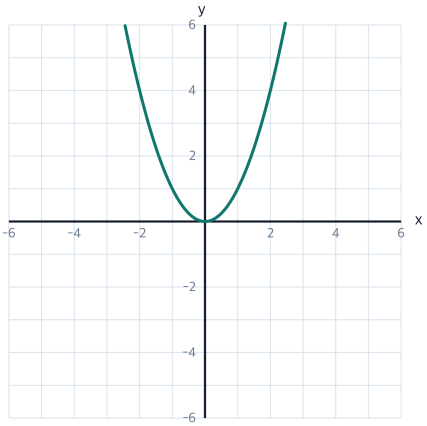
Graph Transformations and Curve Sketching

Parent Graphs

Name _____

SB1

Use the graph to state the requested key features.



For the graph shown, state the domain and range of $f(x) = x^2$.

ALG-146D-L1 · SB

Show your working

KAIZEN MATHS

Lesson starter

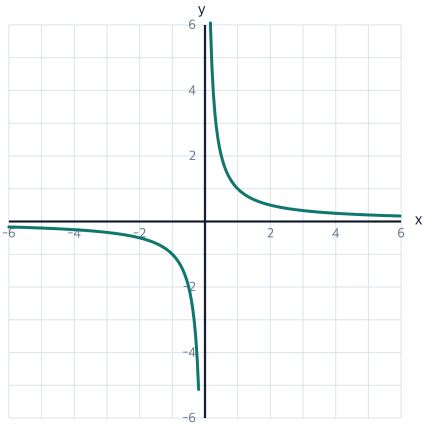
Graph Transformations and Curve Sketching

Parent Graphs

Name _____

SC1

Match each graph to the correct equation.



Match the graph to the correct equation.

A: $f(x) = \frac{1}{x}$

B: $f(x) = |x|$

C: $f(x) = x^3$

Show your working

KAIZEN MATHS

Lesson starter

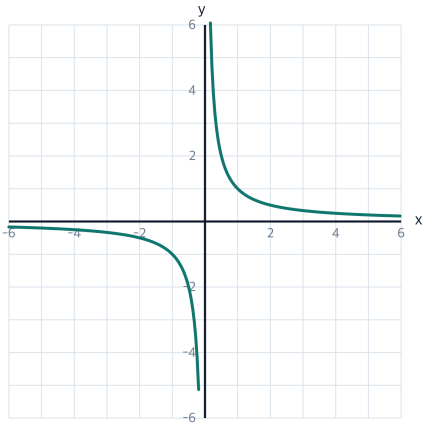
Graph Transformations and Curve Sketching

Parent Graphs

Name _____

SD1

Identify each parent graph and state one key feature.



Identify the parent graph shown and state one key feature.

Show your working

KAIZEN MATHS

Exit ticket

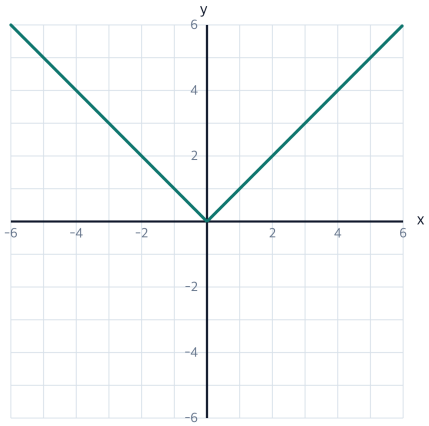
Graph Transformations and Curve Sketching

Parent Graphs

Name _____

EA1

Use the graph to state the requested key features.



For the graph shown, state the domain and range of f(x) = |x|.

ALG-146D-L1 · EA

Show your working

KAIZEN MATHS

Exit ticket

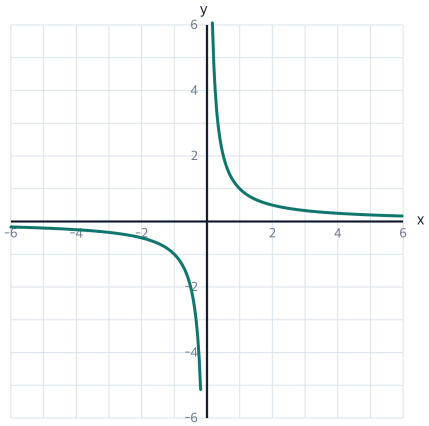
Graph Transformations and Curve Sketching

Parent Graphs

Name _____

EB1

Match each graph to the correct equation.



Match the graph to the correct equation.

A: $f(x) = \frac{1}{x}$

B: $f(x) = 2^x$

C: $f(x) = x^3$

ALG-146D-L1 · EB

Show your working

KAIZEN MATHS

Exit ticket

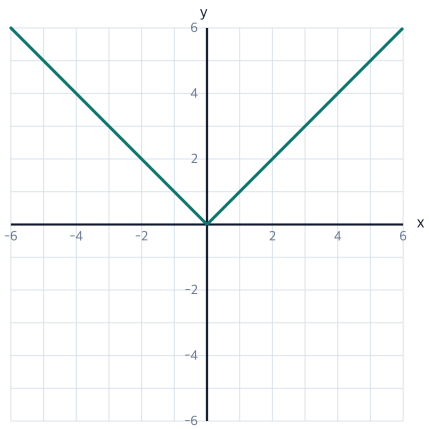
Graph Transformations and Curve Sketching

Parent Graphs

Name _____

EC1

Identify each parent graph and state one key feature.



Identify the parent graph shown and state one key feature.

ALG-146D-L1 · EC

Show your working

KAIZEN MATHS

Exit ticket

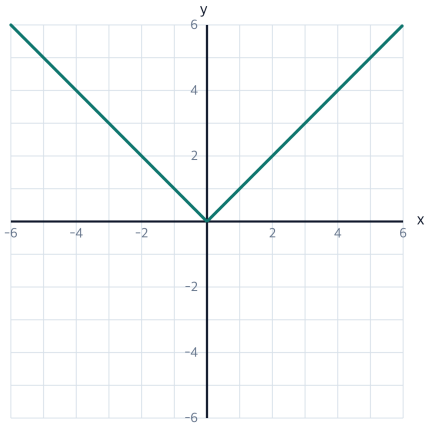
Graph Transformations and Curve Sketching

Parent Graphs

Name _____

EA1

Use the graph to state the requested key features.



For the graph shown, state the domain and range of f(x) = |x|.

ALG-146D-L1 · EA

Show your working

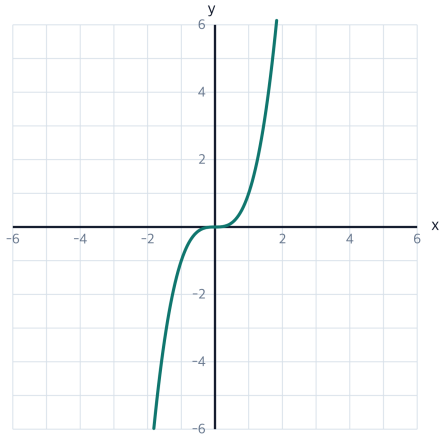
Teacher answers & working

Graph Transformations and Curve Sketching · Parent Graphs

ALG-146D-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Identify each parent graph and state one key feature.



Identify the parent graph shown and state one key feature.

Answer:

$f(x) = x^3$; It passes through the origin and is increasing for all x.

Working:

Look at the overall shape of the graph before choosing an equation.

This shape matches the cubic parent graph: $f(x) = x^3$.

A key feature is: It passes through the origin and is increasing for all x.

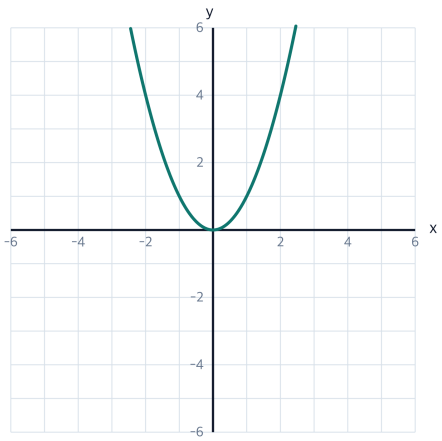
Teacher answers & working

Graph Transformations and Curve Sketching · Parent Graphs

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SB1

Use the graph to state the requested key features.



For the graph shown, state the domain and range of $f(x) = x^2$.

Answer:

Domain: all real values. Range: $y \geq 0$.

Working:

The domain is the set of possible input values on the x-axis.

For $f(x) = x^2$, the domain is all real values.

The range is the set of possible output values on the y-axis.

For this graph, the range is $y \geq 0$.

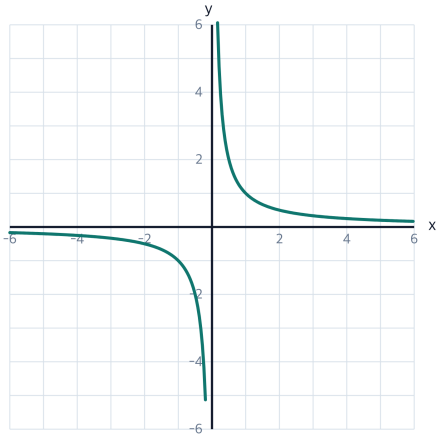
Teacher answers & working

Graph Transformations and Curve Sketching · Parent Graphs

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SC1

Match each graph to the correct equation.



Match the graph to the correct equation.

A: f(x) = $\frac{1}{x}$

B: f(x) = |x|

C: f(x) = x^3

Answer:

A: $f(x) = 1/x$

Working:

Compare the visible graph shape with each option.

The graph has the shape of f(x) = $\frac{1}{x}$.

Therefore the correct match is A: f(x) = 1/x.

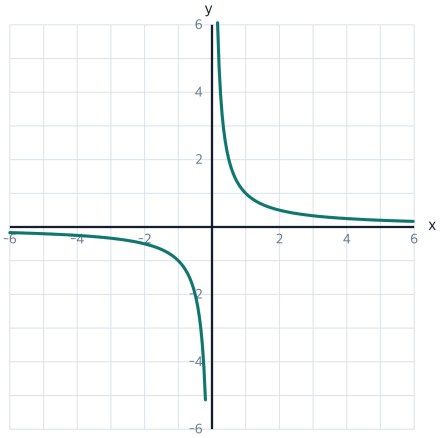
Teacher answers & working

Graph Transformations and Curve Sketching · Parent Graphs

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SD1

Identify each parent graph and state one key feature.



Identify the parent graph shown and state one key feature.

Answer:

f(x) = 1/x; It has asymptotes x = 0 and y = 0.

Working:

Look at the overall shape of the graph before choosing an equation.

This shape matches the reciprocal parent graph: f(x) = $\frac{1}{x}$.

A key feature is: It has asymptotes x = 0 and y = 0.

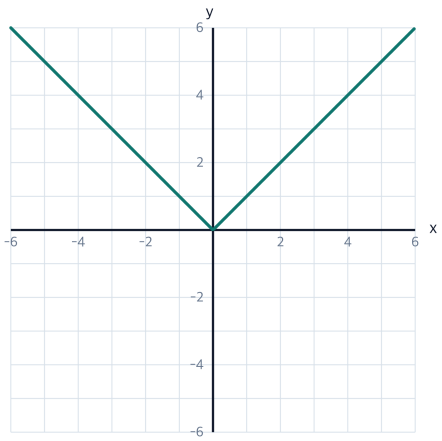
Teacher answers & working

Graph Transformations and Curve Sketching · Parent Graphs

ALG-146D-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Use the graph to state the requested key features.



For the graph shown, state the domain and range of f(x) = |x|.

Answer:

Domain: all real values. Range: y ≥ 0.

Working:

The domain is the set of possible input values on the x-axis.

For f(x) = |x|, the domain is all real values.

The range is the set of possible output values on the y-axis.

For this graph, the range is y ≥ 0.

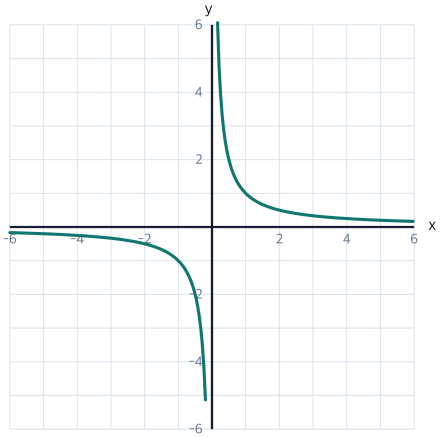
Teacher answers & working

Graph Transformations and Curve Sketching · Parent Graphs

ALG-146D-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

Match each graph to the correct equation.



Match the graph to the correct equation.

A: f(x) = $\frac{1}{x}$

B: f(x) = 2^x

C: f(x) = x^3

Answer:

A: $f(x) = 1/x$

Working:

Compare the visible graph shape with each option.

The graph has the shape of f(x) = $\frac{1}{x}$.

Therefore the correct match is A: f(x) = 1/x.

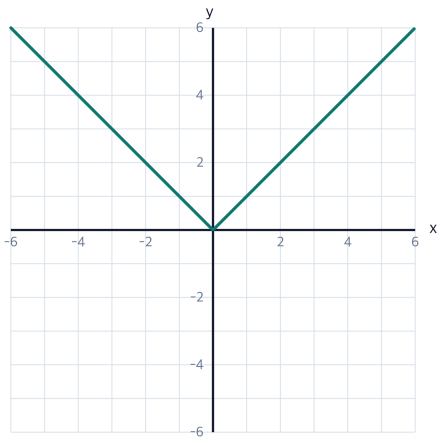
Teacher answers & working

Graph Transformations and Curve Sketching · Parent Graphs

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EC1

Identify each parent graph and state one key feature.



Identify the parent graph shown and state one key feature.

Answer:

f(x) = |x|; It has a sharp minimum point at (0, 0).

Working:

Look at the overall shape of the graph before choosing an equation.

This shape matches the modulus parent graph: f(x) = |x|.

A key feature is: It has a sharp minimum point at (0, 0).